

electronics manufacturing services (EMS) market in India is growing and they too have a large requirement of SMD components."

V.S. Srikanth, chief executive, S.B. Technologies (an EMS provider), foresees a large demand for SMD components in the automotive sector because of miniaturisation. Also, automobiles are gradually incorporating more and more electronic components and systems.

SMD LEDs, in particular, have a wide requirement starting from the consumer market to the industrial sector. Almost all the fields are adopting SMD LEDs because of their durability and size. To name a few, there are areas such as industrial lighting, solar lighting, automation, handheld devices, medical and power sector.

SMD technology is also extending to the outdoor display market. An SMD pixel consists of red, green and blue diodes, mounted in a single package, which is then mounted on the driver PC board. Smaller than a pinhead, the individual diodes are placed very close together. Compared to the discrete diode screen with the same resolution, this arrangement reduces the minimum viewing distance by 25 per cent.

However, most of the demand for SMD LEDs is coming from the lighting industry, as organisations switch to LED lighting systems that are highly energy-efficient. So, in the near future there will be more consumption of SMD LEDs.

Multilayer ceramic chip capacitors (MLCCs) are replacing aluminium electrolytic capacitors in many products. MLCCs make up 62 per cent in the global capacitor market, with the largest shipment and the highest output value. On an average, some 300-400 pieces of MLCCs are needed for each cellphone; some 400-600 pieces of MLCCs are used in each notebook PC; and some 300-400 pieces are required for each tablet PC—according to a report by Reportlinker.

With the expanding MLCC market capacity, a part of aluminium

## Some SMD Component Manufacturers in India

| Manufacturer                         | Location   | Product manufactured                                                            |
|--------------------------------------|------------|---------------------------------------------------------------------------------|
| Gujarat Poly-AVX Electronics         | Mumbai     | Capacitors                                                                      |
| Hi-Tech Resistors                    | Nagpur     | Surface-mount power resistors                                                   |
| Integrated Electronics Technology    | Thane      | Thin-film SMD chip resistors and high-ohmic-value thick-film SMD chip resistors |
| Telecommunication Components And Equ | Aurangabad | Thermistors (SMDs)                                                              |

electrolytic capacitor market is seized by MLCC market. MLCC is a kind of capacitor with the strongest growth momentum, while the aluminium electrolytic capacitor market is expected to shrink in the future.

The aluminium electrolytic capacitor market mainly caters to industrial equipment, white household appliances, TVs and PC motherboards. In particular, the application of aluminium electrolytic capacitor in white household appliances and TVs remains on edge or slips down, while the use in desktop and traditional PC market is on the decline of varying degrees. Since it is ultra-thin, Ultrabook has no choice but to employ MLCC.

### Why manufacturing not picking up

"If I am not mistaken, there is no SMD component manufacturing capability in India worth talking of, the reason being it is a very capital-intensive industry and you need to keep modernising the machinery to keep up with the technology. The duty structure and the rules and regulations in India are so lopsided that there is no attraction for big players to set up shop in India. Import is therefore the preferred route," says Soparkar.

"Putting up an SMD manufacturing plant requires a lot of capital. The economy of scale dictates its capacity, which also has to be very large. If a manufacturer were to put up an SMD plant in India, he would have to tap the export market to be viable. Then, again due to a high duty structure and so many complexities in doing business in India, it is unlikely to be profitable. This is especially so for commercial-grade components. In case

### SMT component basics

Unlike their counterpart through-hole components, SMT components are directly mounted onto a printed circuit board (PCB). In through-hole, pins are inserted into drilled holes on a PCB. SMT components have small metallic end caps, which are directly soldered onto the PCB surface. This technology reduces the package size and improves functionality to a great extent on the same board-area. SMT places components on both sides of the PCB, which helps to reduce time in assembling the components. Placing components on both sides of the board also allows much higher circuit densities.

Surface-mount devices are very small and thin, compared to through-hole. Specifically speaking, these are one-quarter to one-tenth the size and weight, and one-half to one-quarter the cost of equivalent through-hole parts.

This allows lower resistance and inductance at the connection, leading to better performance for high-frequency parts.

Passive SMT components include resistors and capacitors, while active SMT components include diodes, transistors and integrated circuits.

of specialised high-value components, it may be viable if we have the latest or niche technology available," Soparkar explains.

"We have lost the opportunity to make components in India. There is absolutely no government support, but only delays in each and every policy. No one bothers about the sorry state of component manufacture in India. The government and trade/manufacturers' associations have to work together. The government is only interested in collecting taxes,"